

EXCEL

FILTER

© ICT Society ℗ Kapasa Makasa University | Prepared by James Soko

INTRODUCTION

- The FILTER function in Excel is used to return specific data from a range based on one or more conditions. It helps extract relevant data dynamically without manually sorting or searching.
- Can be used in Fraud Prevention & Detection Technologies to;

1. Identifying Malicious IPs Quickly
2. Analyzing Fraudulent Transactions
3. Tracking Phishing Attacks
4. Real-Time Monitoring & Incident Response

SYNTAX OF FILTER

- =FILTER(array, include, [if_empty])

1. array → The range of data you want to filter.
2. include → The condition(s) used to filter the data.
3. if_empty (optional) → The value to return if no data matches the condition.

© ICT Society ℗ Kapasa Makasa University | Prepared by James Soko

USAGE

1. Filter Based on a Single Condition

- =FILTER(A4:E13,C4:C13="High","Invalid Severity Level")
- This will return only rows where Threat Level (column C) is "High“

2. Filter Based on Multiple Conditions

- =FILTER(A4:E13, (B4:B13="Malicious") * (C4:C13="High"), "Invalid Entry")
- The * acts as AND logic, meaning both conditions must be met.

3. Filter Data Dynamically

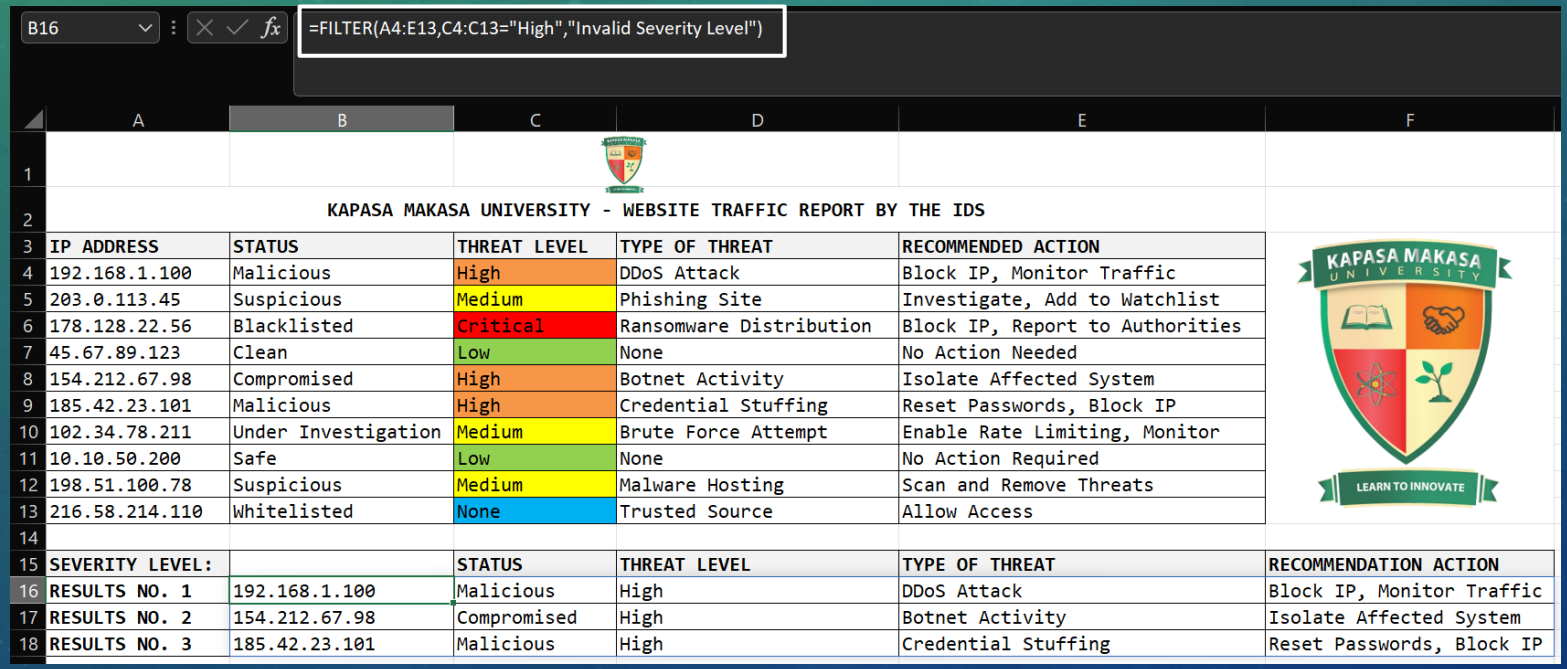
- =FILTER(A4:E13,C4:C13=B15,"Invalid Risk Level")
- This will update results automatically as the value in G1 changes.

APPLICATION

	A	B	C	D	E	F
1						
2	KAPASA MAKASA UNIVERSITY - WEBSITE TRAFFIC REPORT BY THE IDS					
3	IP ADDRESS	STATUS	THREAT LEVEL	TYPE OF THREAT	RECOMMENDED ACTION	
4	192.168.1.100	Malicious	High	DDoS Attack	Block IP, Monitor Traffic	
5	203.0.113.45	Suspicious	Medium	Phishing Site	Investigate, Add to Watchlist	
6	178.128.22.56	Blacklisted	Critical	Ransomware Distribution	Block IP, Report to Authorities	
7	45.67.89.123	Clean	Low	None	No Action Needed	
8	154.212.67.98	Compromised	High	Botnet Activity	Isolate Affected System	
9	185.42.23.101	Malicious	High	Credential Stuffing	Reset Passwords, Block IP	
10	102.34.78.211	Under Investigation	Medium	Brute Force Attempt	Enable Rate Limiting, Monitor	
11	10.10.50.200	Safe	Low	None	No Action Required	
12	198.51.100.78	Suspicious	Medium	Malware Hosting	Scan and Remove Threats	
13	216.58.214.110	Whitelisted	None	Trusted Source	Allow Access	
14						
15	SEVERITY LEVEL:		STATUS	THREAT LEVEL	TYPE OF THREAT	RECOMMENDATION ACTION
16	RESULTS NO. 1					
17	RESULTS NO. 2					
18	RESULTS NO. 3					

FILTER BASED ON SINGLE CONDITION

- In cell B16 (Formula Bar) Enter: =FILTER(A4:E13,C4:C13="High","Invalid Severity Level")



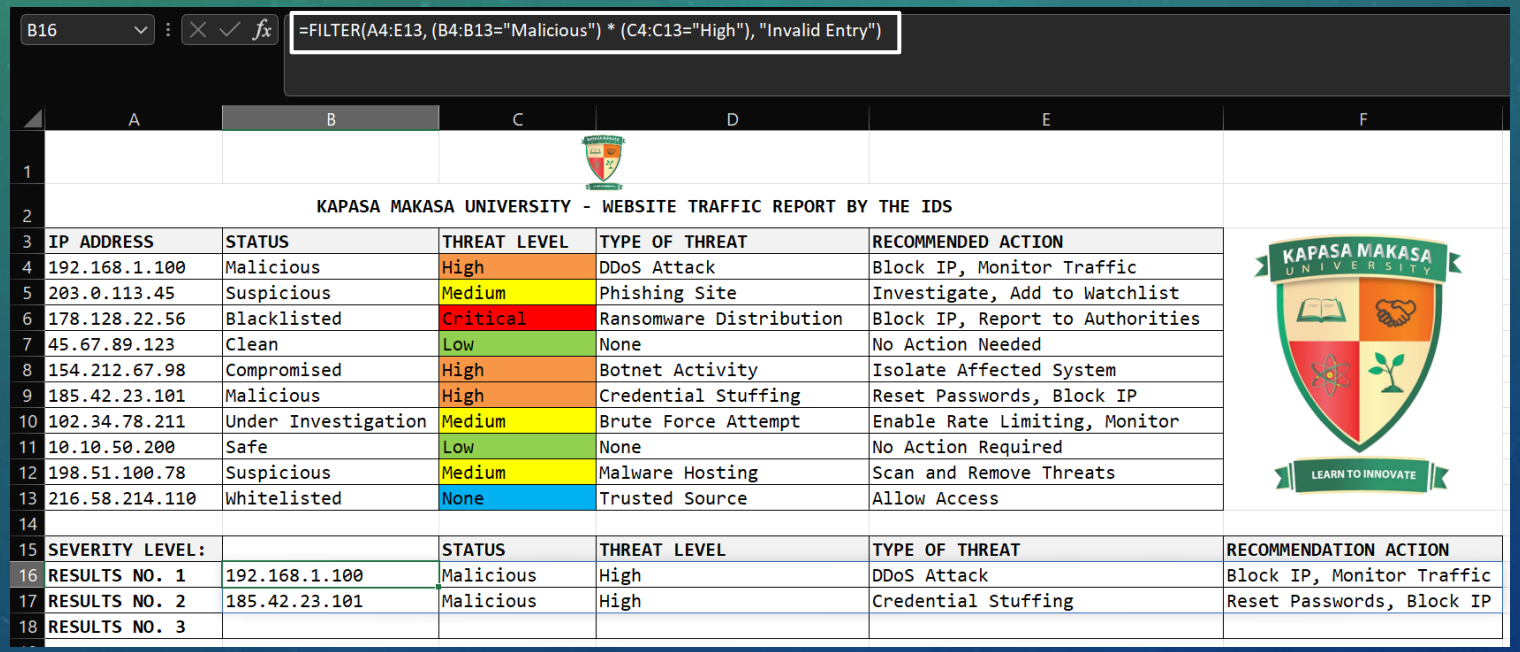
KAPASA MAKASA UNIVERSITY - WEBSITE TRAFFIC REPORT BY THE IDS

IP ADDRESS	STATUS	THREAT LEVEL	TYPE OF THREAT	RECOMMENDED ACTION
192.168.1.100	Malicious	High	DDoS Attack	Block IP, Monitor Traffic
203.0.113.45	Suspicious	Medium	Phishing Site	Investigate, Add to Watchlist
178.128.22.56	Blacklisted	Critical	Ransomware Distribution	Block IP, Report to Authorities
45.67.89.123	Clean	Low	None	No Action Needed
154.212.67.98	Compromised	High	Botnet Activity	Isolate Affected System
185.42.23.101	Malicious	High	Credential Stuffing	Reset Passwords, Block IP
102.34.78.211	Under Investigation	Medium	Brute Force Attempt	Enable Rate Limiting, Monitor
10.10.50.200	Safe	Low	None	No Action Required
198.51.100.78	Suspicious	Medium	Malware Hosting	Scan and Remove Threats
216.58.214.110	Whitelisted	None	Trusted Source	Allow Access

SEVERITY LEVEL:		STATUS	THREAT LEVEL	TYPE OF THREAT	RECOMMENDATION ACTION
RESULTS NO. 1	192.168.1.100	Malicious	High	DDoS Attack	Block IP, Monitor Traffic
RESULTS NO. 2	154.212.67.98	Compromised	High	Botnet Activity	Isolate Affected System
RESULTS NO. 3	185.42.23.101	Malicious	High	Credential Stuffing	Reset Passwords, Block IP

FILTER BASED ON MULTIPLE CONDITIONS

- In cell B16 (Formula Bar) Enter: =FILTER(A4:E13, (B4:B13="Malicious") * (C4:C13="High"), "Invalid Entry")



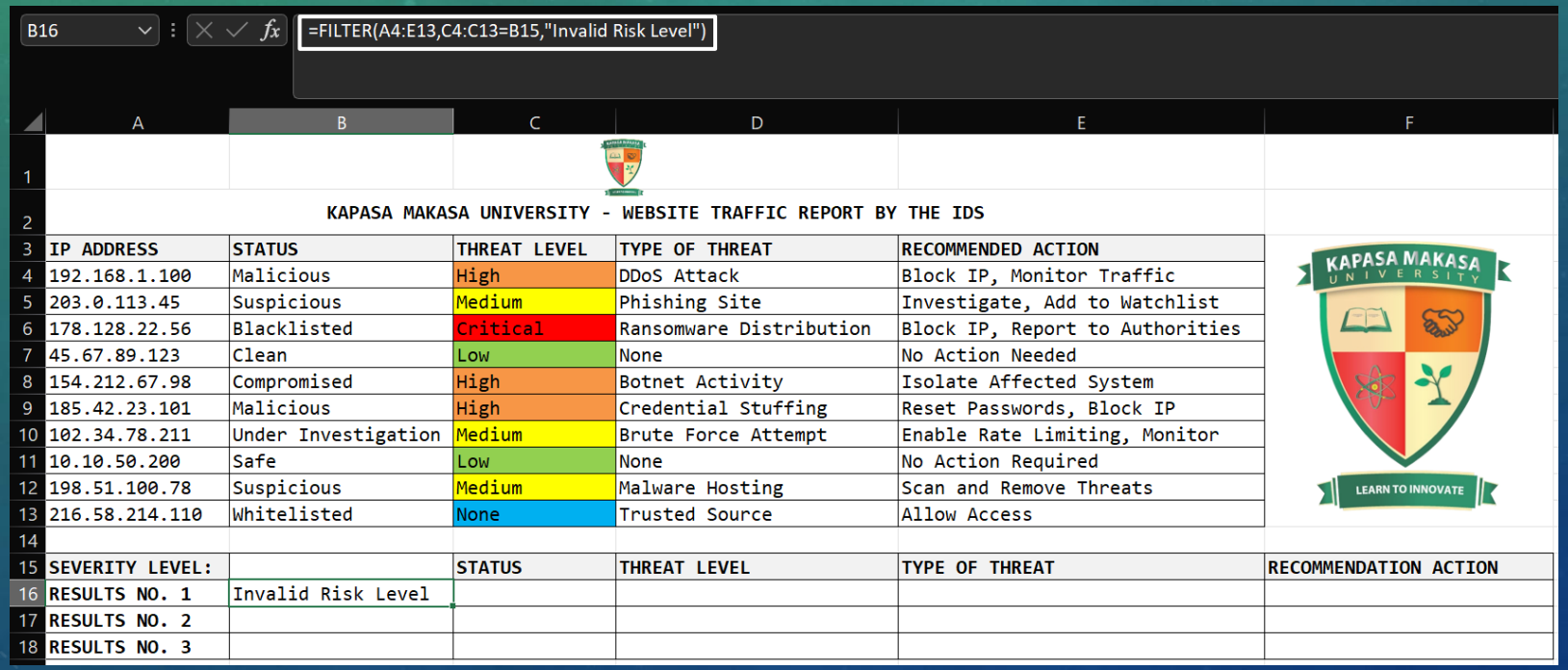
KAPASA MAKASA UNIVERSITY - WEBSITE TRAFFIC REPORT BY THE IDS

IP ADDRESS	STATUS	THREAT LEVEL	TYPE OF THREAT	RECOMMENDED ACTION
192.168.1.100	Malicious	High	DDoS Attack	Block IP, Monitor Traffic
203.0.113.45	Suspicious	Medium	Phishing Site	Investigate, Add to Watchlist
178.128.22.56	Blacklisted	Critical	Ransomware Distribution	Block IP, Report to Authorities
45.67.89.123	Clean	Low	None	No Action Needed
154.212.67.98	Compromised	High	Botnet Activity	Isolate Affected System
185.42.23.101	Malicious	High	Credential Stuffing	Reset Passwords, Block IP
102.34.78.211	Under Investigation	Medium	Brute Force Attempt	Enable Rate Limiting, Monitor
10.10.50.200	Safe	Low	None	No Action Required
198.51.100.78	Suspicious	Medium	Malware Hosting	Scan and Remove Threats
216.58.214.110	Whitelisted	None	Trusted Source	Allow Access

SEVERITY LEVEL:		STATUS	THREAT LEVEL	TYPE OF THREAT	RECOMMENDATION ACTION
RESULTS NO. 1	192.168.1.100	Malicious	High	DDoS Attack	Block IP, Monitor Traffic
RESULTS NO. 2	185.42.23.101	Malicious	High	Credential Stuffing	Reset Passwords, Block IP
RESULTS NO. 3					

FILTER DATA DYNAMICALLY

- In cell B16 (Formula Bar) Enter: =FILTER(A4:E13,C4:C13=B15,"Invalid Risk Level")



B16 =FILTER(A4:E13,C4:C13=B15,"Invalid Risk Level")

	A	B	C	D	E	F
1						
2	KAPASA MAKASA UNIVERSITY - WEBSITE TRAFFIC REPORT BY THE IDS					
3	IP ADDRESS	STATUS	THREAT LEVEL	TYPE OF THREAT	RECOMMENDED ACTION	
4	192.168.1.100	Malicious	High	DDoS Attack	Block IP, Monitor Traffic	
5	203.0.113.45	Suspicious	Medium	Phishing Site	Investigate, Add to Watchlist	
6	178.128.22.56	Blacklisted	Critical	Ransomware Distribution	Block IP, Report to Authorities	
7	45.67.89.123	Clean	Low	None	No Action Needed	
8	154.212.67.98	Compromised	High	Botnet Activity	Isolate Affected System	
9	185.42.23.101	Malicious	High	Credential Stuffing	Reset Passwords, Block IP	
10	102.34.78.211	Under Investigation	Medium	Brute Force Attempt	Enable Rate Limiting, Monitor	
11	10.10.50.200	Safe	Low	None	No Action Required	
12	198.51.100.78	Suspicious	Medium	Malware Hosting	Scan and Remove Threats	
13	216.58.214.110	Whitelisted	None	Trusted Source	Allow Access	
14						
15	SEVERITY LEVEL:		STATUS	THREAT LEVEL	TYPE OF THREAT	RECOMMENDATION ACTION
16	RESULTS NO. 1	Invalid Risk Level				
17	RESULTS NO. 2					
18	RESULTS NO. 3					

FILTER DATA DYNAMICALLY (CONTINUES…)

- Type “None” in cell B15 (Formula Bar) and press Enter.

	A	B	C	D	E	F
1						
2	KAPASA MAKASA UNIVERSITY - WEBSITE TRAFFIC REPORT BY THE IDS					
3	IP ADDRESS	STATUS	THREAT LEVEL	TYPE OF THREAT	RECOMMENDED ACTION	
4	192.168.1.100	Malicious	High	DDoS Attack	Block IP, Monitor Traffic	
5	203.0.113.45	Suspicious	Medium	Phishing Site	Investigate, Add to Watchlist	
6	178.128.22.56	Blacklisted	Critical	Ransomware Distribution	Block IP, Report to Authorities	
7	45.67.89.123	Clean	Low	None	No Action Needed	
8	154.212.67.98	Compromised	High	Botnet Activity	Isolate Affected System	
9	185.42.23.101	Malicious	High	Credential Stuffing	Reset Passwords, Block IP	
10	102.34.78.211	Under Investigation	Medium	Brute Force Attempt	Enable Rate Limiting, Monitor	
11	10.10.50.200	Safe	Low	None	No Action Required	
12	198.51.100.78	Suspicious	Medium	Malware Hosting	Scan and Remove Threats	
13	216.58.214.110	Whitelisted	None	Trusted Source	Allow Access	
14						
15	SEVERITY LEVEL:	None	STATUS	THREAT LEVEL	TYPE OF THREAT	RECOMMENDATION ACTION
16	RESULTS NO. 1	216.58.214.110	Whitelisted	None	Trusted Source	Allow Access
17	RESULTS NO. 2					
18	RESULTS NO. 3					

FILTER DATA DYNAMICALLY (CONTINUES…)

- Type “Low” in cell B15 (Formula Bar) and press Enter.

	A	B	C	D	E	F
1						
2	KAPASA MAKASA UNIVERSITY - WEBSITE TRAFFIC REPORT BY THE IDS					
3	IP ADDRESS	STATUS	THREAT LEVEL	TYPE OF THREAT	RECOMMENDED ACTION	
4	192.168.1.100	Malicious	High	DDoS Attack	Block IP, Monitor Traffic	
5	203.0.113.45	Suspicious	Medium	Phishing Site	Investigate, Add to Watchlist	
6	178.128.22.56	Blacklisted	Critical	Ransomware Distribution	Block IP, Report to Authorities	
7	45.67.89.123	Clean	Low	None	No Action Needed	
8	154.212.67.98	Compromised	High	Botnet Activity	Isolate Affected System	
9	185.42.23.101	Malicious	High	Credential Stuffing	Reset Passwords, Block IP	
10	102.34.78.211	Under Investigation	Medium	Brute Force Attempt	Enable Rate Limiting, Monitor	
11	10.10.50.200	Safe	Low	None	No Action Required	
12	198.51.100.78	Suspicious	Medium	Malware Hosting	Scan and Remove Threats	
13	216.58.214.110	Whitelisted	None	Trusted Source	Allow Access	
14						
15	SEVERITY LEVEL:	Low	STATUS	THREAT LEVEL	TYPE OF THREAT	RECOMMENDATION ACTION
16	RESULTS NO. 1	45.67.89.123	Clean	Low	None	No Action Needed
17	RESULTS NO. 2	10.10.50.200	Safe	Low	None	No Action Required
18	RESULTS NO. 3					

FILTER DATA DYNAMICALLY (CONTINUES…)

- Type “Medium” in cell B15 (Formula Bar) and press Enter.

	A	B	C	D	E	F
1						
2	KAPASA MAKASA UNIVERSITY - WEBSITE TRAFFIC REPORT BY THE IDS					
3	IP ADDRESS	STATUS	THREAT LEVEL	TYPE OF THREAT	RECOMMENDED ACTION	
4	192.168.1.100	Malicious	High	DDoS Attack	Block IP, Monitor Traffic	
5	203.0.113.45	Suspicious	Medium	Phishing Site	Investigate, Add to Watchlist	
6	178.128.22.56	Blacklisted	Critical	Ransomware Distribution	Block IP, Report to Authorities	
7	45.67.89.123	Clean	Low	None	No Action Needed	
8	154.212.67.98	Compromised	High	Botnet Activity	Isolate Affected System	
9	185.42.23.101	Malicious	High	Credential Stuffing	Reset Passwords, Block IP	
10	102.34.78.211	Under Investigation	Medium	Brute Force Attempt	Enable Rate Limiting, Monitor	
11	10.10.50.200	Safe	Low	None	No Action Required	
12	198.51.100.78	Suspicious	Medium	Malware Hosting	Scan and Remove Threats	
13	216.58.214.110	Whitelisted	None	Trusted Source	Allow Access	
14						
15	SEVERITY LEVEL:	Medium	STATUS	THREAT LEVEL	TYPE OF THREAT	RECOMMENDATION ACTION
16	RESULTS NO. 1	203.0.113.45	Suspicious	Medium	Phishing Site	Investigate, Add to Watchlist
17	RESULTS NO. 2	102.34.78.211	Under Investiga	Medium	Brute Force Attempt	Enable Rate Limiting, Monitor
18	RESULTS NO. 3	198.51.100.78	Suspicious	Medium	Malware Hosting	Scan and Remove Threats

FILTER DATA DYNAMICALLY (CONTINUES…)

- Type “High” in cell B15 (Formula Bar) and press Enter.

	A	B	C	D	E	F
1						
2	KAPASA MAKASA UNIVERSITY - WEBSITE TRAFFIC REPORT BY THE IDS					
3	IP ADDRESS	STATUS	THREAT LEVEL	TYPE OF THREAT	RECOMMENDED ACTION	
4	192.168.1.100	Malicious	High	DDoS Attack	Block IP, Monitor Traffic	
5	203.0.113.45	Suspicious	Medium	Phishing Site	Investigate, Add to Watchlist	
6	178.128.22.56	Blacklisted	Critical	Ransomware Distribution	Block IP, Report to Authorities	
7	45.67.89.123	Clean	Low	None	No Action Needed	
8	154.212.67.98	Compromised	High	Botnet Activity	Isolate Affected System	
9	185.42.23.101	Malicious	High	Credential Stuffing	Reset Passwords, Block IP	
10	102.34.78.211	Under Investigation	Medium	Brute Force Attempt	Enable Rate Limiting, Monitor	
11	10.10.50.200	Safe	Low	None	No Action Required	
12	198.51.100.78	Suspicious	Medium	Malware Hosting	Scan and Remove Threats	
13	216.58.214.110	Whitelisted	None	Trusted Source	Allow Access	
14						
15	SEVERITY LEVEL:	High	STATUS	THREAT LEVEL	TYPE OF THREAT	RECOMMENDATION ACTION
16	RESULTS NO. 1	192.168.1.100	Malicious	High	DDoS Attack	Block IP, Monitor Traffic
17	RESULTS NO. 2	154.212.67.98	Compromised	High	Botnet Activity	Isolate Affected System
18	RESULTS NO. 3	185.42.23.101	Malicious	High	Credential Stuffing	Reset Passwords, Block IP

FILTER DATA DYNAMICALLY (CONTINUES…)

- Type “Critical” in cell B15 (Formula Bar) and press Enter.

	A	B	C	D	E	F	G
1							
2	KAPASA MAKASA UNIVERSITY - WEBSITE TRAFFIC REPORT BY THE IDS						
3	IP ADDRESS	STATUS	THREAT LEVEL	TYPE OF THREAT	RECOMMENDED ACTION		
4	192.168.1.100	Malicious	High	DDoS Attack	Block IP, Monitor Traffic		
5	203.0.113.45	Suspicious	Medium	Phishing Site	Investigate, Add to Watchlist		
6	178.128.22.56	Blacklisted	Critical	Ransomware Distribution	Block IP, Report to Authorities		
7	45.67.89.123	Clean	Low	None	No Action Needed		
8	154.212.67.98	Compromised	High	Botnet Activity	Isolate Affected System		
9	185.42.23.101	Malicious	High	Credential Stuffing	Reset Passwords, Block IP		
10	102.34.78.211	Under Investigation	Medium	Brute Force Attempt	Enable Rate Limiting, Monitor		
11	10.10.50.200	Safe	Low	None	No Action Required		
12	198.51.100.78	Suspicious	Medium	Malware Hosting	Scan and Remove Threats		
13	216.58.214.110	Whitelisted	None	Trusted Source	Allow Access		
14							
15	SEVERITY LEVEL:	Critical	STATUS	THREAT LEVEL	TYPE OF THREAT	RECOMMENDATION ACTION	
16	RESULTS NO. 1	178.128.22.56	Blacklisted	Critical	Ransomware Distribution	Block IP, Report to Authorities	
17	RESULTS NO. 2						
18	RESULTS NO. 3						

ICT SOCIETY | EXCEL SLIDES

PREPARED BY JAMES SOKO



© ICT Society ℗ Kapasa Makasa University | Prepared by James Soko